

# 6

## WAVES AND SOUND

### CHAPTER OUTLINE

**6.1** Waves—Types  
and Properties  
**6.2** Aspects of Wave  
Propagation

**6.3** Sound  
**6.4** Production of Sound  
**6.5** Propagation of Sound  
**6.6** Perception of Sound

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# 6.1 Waves—Types and Properties

Waves are many and diverse, but they share some basic features. They all involve *vibrations* or *oscillations* of some kind.

Floating leaves show the vibration of the water's surface as ripples move by. Our ears respond to the oscillation of air molecules and give us the perception of sound.

Also, waves move and *carry energy* yet do not have mass. The sound from a loudspeaker can break a glass even though no matter moves from the speaker to the glass.

## DEFINITION

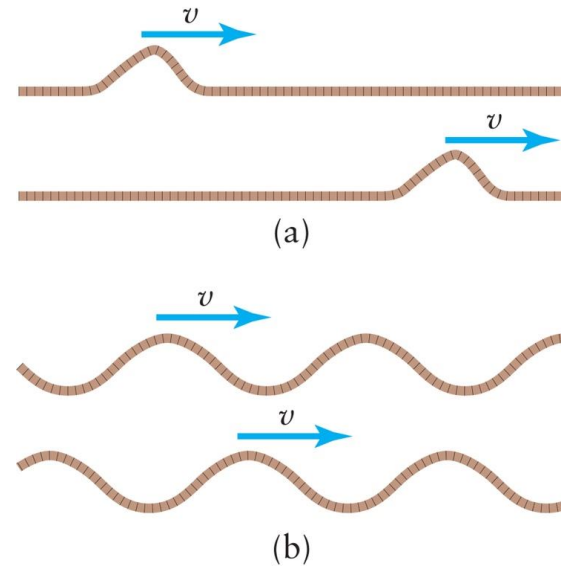
**Wave:** A traveling disturbance consisting of coordinated vibrations that transmit energy with no net movement of matter.

# 6.1 Waves—Types and Properties

- The substance through which such waves travel is called the *medium* of the wave. Particles of the medium vibrate in a coordinated fashion to form the wave.
- Many waves—sound, water ripples, waves on a rope—require a material medium. They cannot exist in a vacuum. On the other hand, light, radio waves, microwaves, and x-rays can travel through a vacuum because they do not require a medium for their propagation.
- Waves occur in a great variety of substances: in gases (sound), liquids (water ripples), and solids (seismic waves through rock).
- Some travel along a line (a wave on a rope), some across a surface (vibrations on a drumhead) and some throughout space in three dimensions (light).

# 6.1 Waves—Types and Properties

- A wave can be short and fleeting, called a *wave pulse*, or steady and repeating, called a *continuous wave*.
- The sound of a bursting balloon, a tsunami (large ocean wave generated by an earthquake), and the light from a camera flash are examples of wave pulses.
- The sound from a tuning fork and the light from the Sun are continuous waves.



# 6.1 Waves—Types and Properties

There are two main wave types: **transverse** and **longitudinal**.

## DEFINITION

**Transverse Wave:** A wave in which the oscillations are perpendicular (transverse) to the direction the wave travels. Examples: waves on a rope, electromagnetic waves, some seismic waves.

**Longitudinal Wave:** A wave in which the oscillations are along the direction the wave travels. Examples: sound in the air, some seismic waves.

## 6.1 Waves—Types and Properties

The **speed of a wave** is the rate of movement of the disturbance. For a given type of wave, the speed is determined by the properties of the medium.

For example, the speed of waves on a stretched rope or a Slinky can be computed by using the force  **$F$**  that must be exerted to keep it stretched and its linear mass density  **$\rho$** , which equals its mass  **$m$**  divided by its length  **$l$** . The speed at which a wave travels on the string when it is under a tension  **$T$**  is

$$v = \sqrt{\frac{F}{\rho}} \quad \left( \text{wave on a rope or spring, } \rho = \frac{m}{l} \right)$$

# 6.1 Waves—Types and Properties

From this we see that the speed:

Increases as the tension increases

- The string has a greater restoring force that attempts to straighten it out.

Is faster for smaller strings

- The string has less mass that has to be moved by the restoring force.

Is independent of the length

- The speed depends on the mass per length, not on just the length.

## Example 6.1

A student stretches a Slinky out on the floor to a length of 2 meters. The force needed to keep the Slinky stretched is measured and found to be 1.2 newtons. The Slinky's mass is 0.3 kilograms. What is the speed of any wave sent down the Slinky by the student?

$$\rho = \frac{m}{l} = \frac{0.3 \text{ kg}}{2 \text{ m}} = 0.15 \text{ kg/m}$$

$$\begin{aligned} v &= \sqrt{\frac{F}{\rho}} = \sqrt{\frac{1.2 \text{ N}}{0.15 \text{ kg/m}}} \\ &= \sqrt{8 \text{ m}^2/\text{s}^2} = 2.8 \text{ m/s} \end{aligned}$$



## 6.1 Waves—Types and Properties

- The speed of sound in air or any other gas depends on the ratio of the pressure of the gas to the density of the gas.
  - But for each gas, this ratio depends only on the temperature.
- In particular, the speed of sound in a gas is proportional to the square root of the Kelvin temperature:

$$v \propto \sqrt{T}$$

( $v$  in m/s,  $T$  in kelvins)

## 6.1 Waves—Types and Properties

- The speed of a sound wave when the air is at a temperature  $T$  is

$$v = 20.1 \times \sqrt{T}.$$

The temperature must be in Kelvin.

- For other gases

Speed of sound in helium:  $v = 58.8 \times \sqrt{T}$  (SI units)

Speed of sound in carbon dioxide ( $\text{CO}_2$ ):  $v = 1.57 \times \sqrt{T}$  (SI units)

## Example 6.2

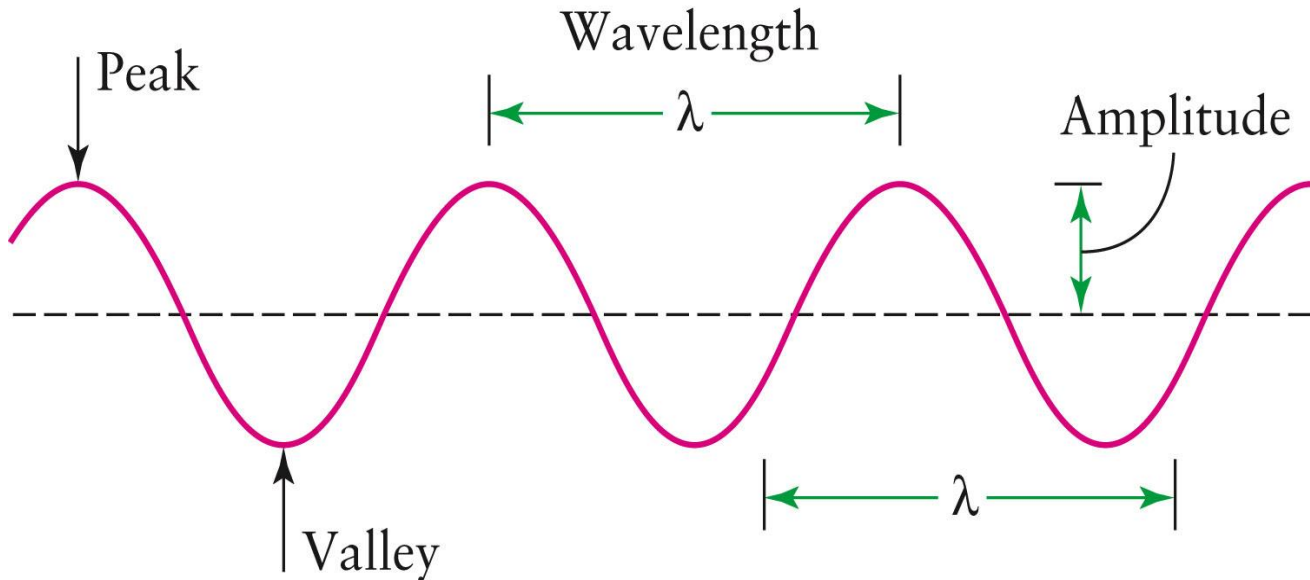
What is the speed of sound in air at room temperature ( $20^{\circ}\text{C} = 68^{\circ}\text{F}$ )?

$$T = 273 + 20 = 293 \text{ K}$$

$$\begin{aligned} v &= 20.1 \times \sqrt{T} = 20.1 \times \sqrt{293} = 20.1 \times 17.1 \\ &= 344 \text{ m/s (770 mph)} \end{aligned}$$

# 6.1 Waves—Types and Properties: Amplitude, Wavelength, and Frequency

- The figure shows a “snapshot” of a continuous wave.
- In addition to **wave speed**, there are three other important parameters of a continuous wave that can be measured: **amplitude**, **wavelength**, and **frequency**



# 6.1 Waves—Types and Properties: Amplitude, Wavelength, and Frequency

## DEFINITIONS

**Amplitude:** The maximum displacement of points on a wave measured from the equilibrium position.

**Wavelength:** The distance between two successive “like” points on a wave. For example, the distance between two adjacent peaks or two adjacent valleys. Wavelength is represented by the lowercase Greek letter lambda ( $\lambda$ ).

**Frequency:** The number of cycles of a wave passing a point per unit time. The number of oscillations per second in the wave.

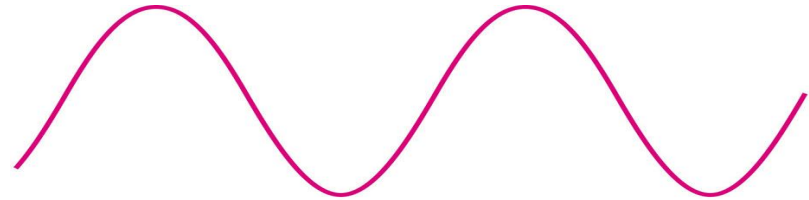
## 6.1 Waves—Types and Properties: Amplitude, Wavelength, and Frequency

- Any segment of a wave that is one wavelength long is called one *cycle* of the wave.
- Amplitude and wavelength are independent features of a wave: a short wavelength wave can have a small or a large amplitude.
- Amplitude, wavelength, and frequency can be identified for both transverse waves and longitudinal waves.

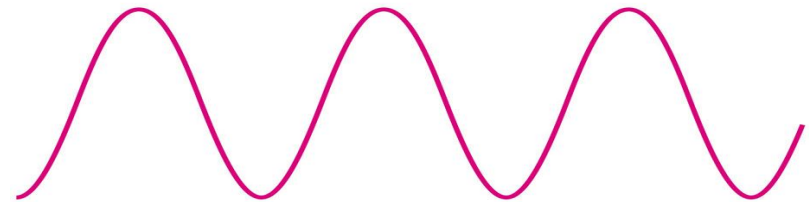
# 6.1 Waves—Types and Properties: Amplitude, Wavelength, and Frequency

Here is an illustration of changing the wavelength and/or amplitude.

- Lower amplitude implies smaller height/depth
- Shorter wavelength implies more complete waves “fit” in a given distance



Original wave



Shorter wavelength, same amplitude



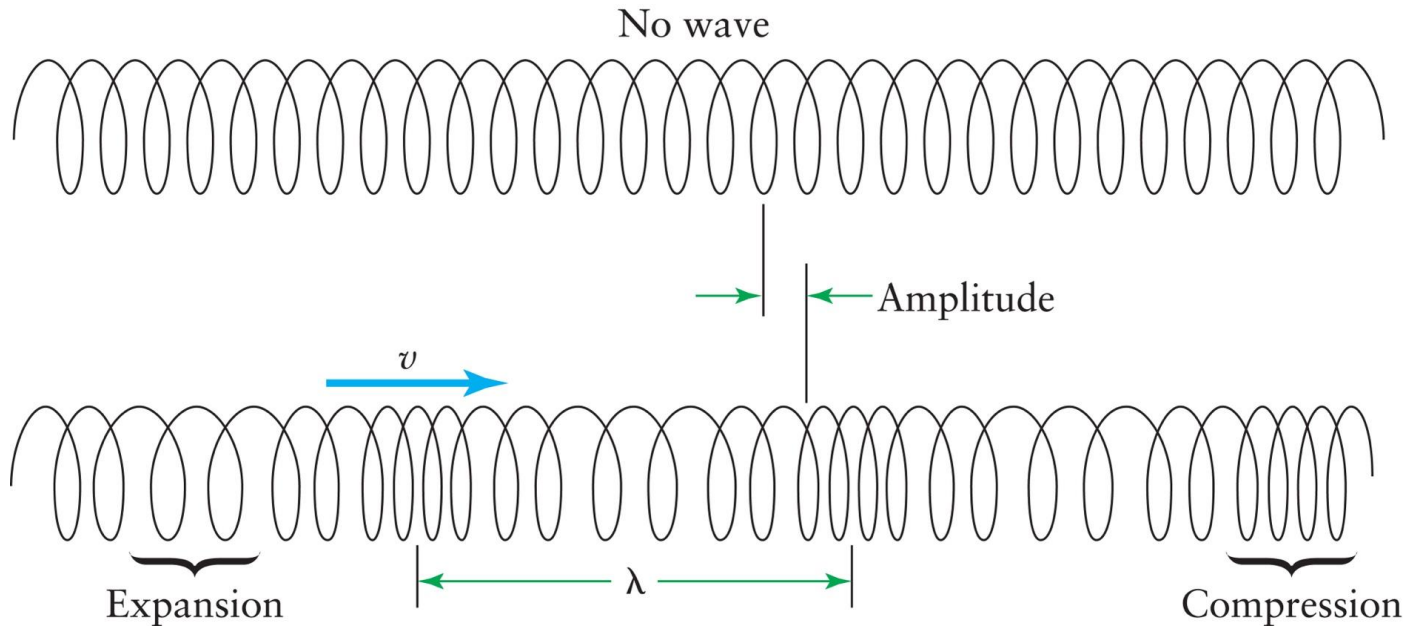
Same wavelength, smaller amplitude



Shorter wavelength, smaller amplitude

# 6.1 Waves—Types and Properties: Amplitude, Wavelength, and Frequency

- We use different terminology for the “peaks” and “valleys” of a longitudinal wave.
  - A **compression** is where the medium is squeezed together.
  - A **expansion** is where the medium is spread apart.



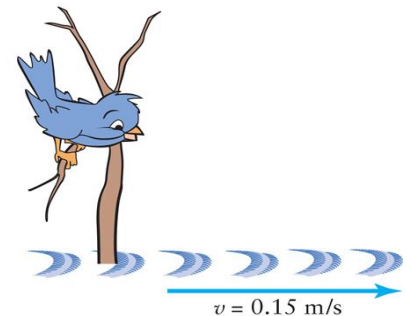
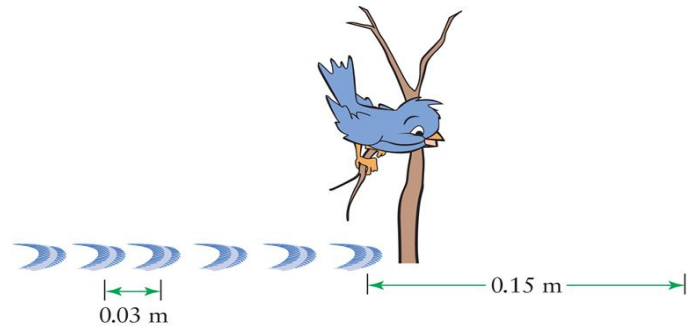


# 6.1 Waves—Types and Properties: Amplitude, Wavelength, and Frequency

- The speed of a wave, its wavelength, and its frequency are related to each other in a simple way. Imagine a continuous wave passing by a point, perhaps ripples moving by a plant stem. The speed of the wave equals the number of cycles that pass by each second multiplied by the length of each cycle.

$$v = f\lambda$$

- For example, if five cycles pass the stem each second and the peaks of the ripples are 0.03 meters apart, the wave speed is 0.15 m/s.



## 6.1 Waves—Types and Properties: Amplitude, Wavelength, and Frequency

- In many cases, all waves that travel in a particular medium have the same speed. Sound is an important example of this; sound pulses, low frequency sounds, and high-frequency sounds travel through the air with the same speed: 344 m/s at room temperature. Similarly, light, radio waves, and microwaves travel with the same speed in a vacuum:  $3 \times 10^8$  m/s.
- According to the equation  $v = f \lambda$ , when the wave speed is the same for all waves, *higher* frequency waves must have proportionally *shorter* wavelengths.

## Example 6.3

Before a concert, musicians in an orchestra tune their instruments to the note A, which has a frequency of 440 hertz. What is the wavelength of this sound in air at room temperature?

$$v = f \lambda$$

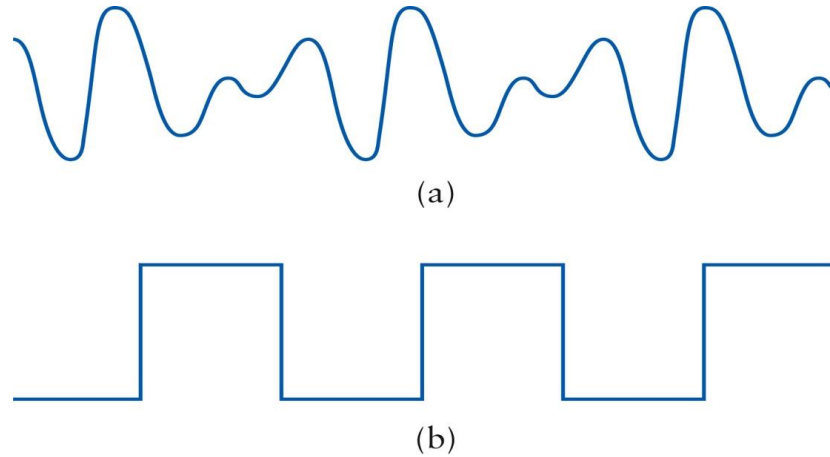
$$344 \text{ m/s} = 440 \text{ Hz} \times \lambda$$

$$\frac{344 \text{ m/s}}{440 \text{ Hz}} = \lambda$$

$$\lambda = 0.78 \text{ m} = 2.6 \text{ ft}$$

# 6.1 Waves—Types and Properties: Amplitude, Wavelength, and Frequency

- Any continuous wave that does not have a sinusoidal shape is called a *complex wave*.
- A *complex wave* is any continuous wave that does not have a sinusoidal shape.
- The shape of a wave is called its *waveform*.
- These two complex waves in the figure have about the same wavelength and amplitude, but they have very different waveforms.

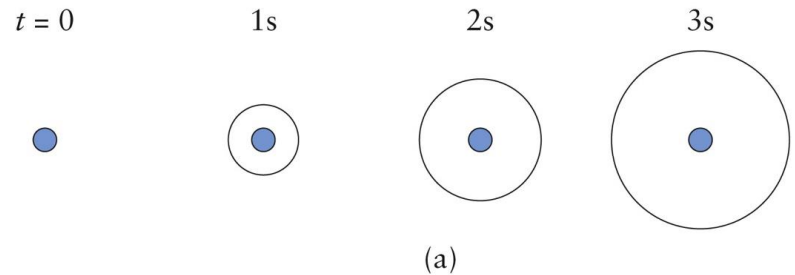


## 6.2 Aspects of Wave Propagation: Wave Fronts and Rays

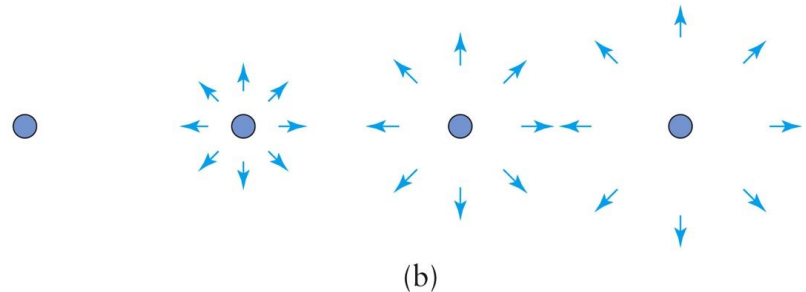
In this section, we consider what waves do as they travel. There are two approaches to represent waves traveling along a surface or throughout space in three dimensions.

In the figure:

(a) A wave front shows how a pulse spreads over water. The wavefront is a circle representing the location of a wave peak.

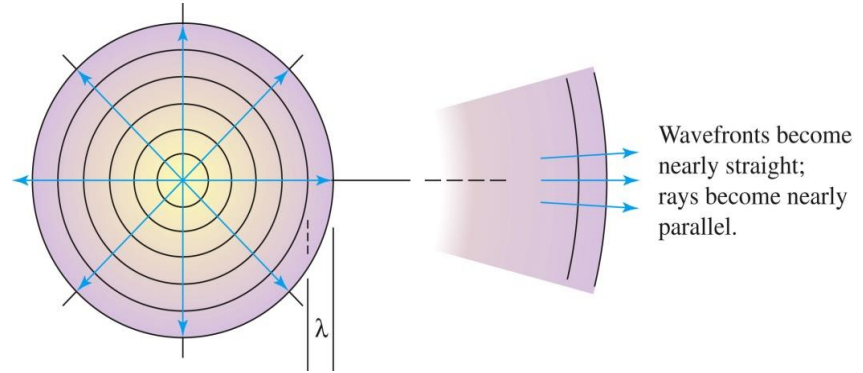


(b) The same wave pulse, at the same times, represented with rays. A ray is an arrow representing the direction that a wave segment is traveling.



## 6.2 Aspects of Wave Propagation: Wave Fronts and Rays

- For a continuous water wave, the wave fronts are concentric circles around the point of origin (the “source” of the wave) that represent individual peaks of the wave.
- a continuous wave is like a series of wave pulses produced one after another

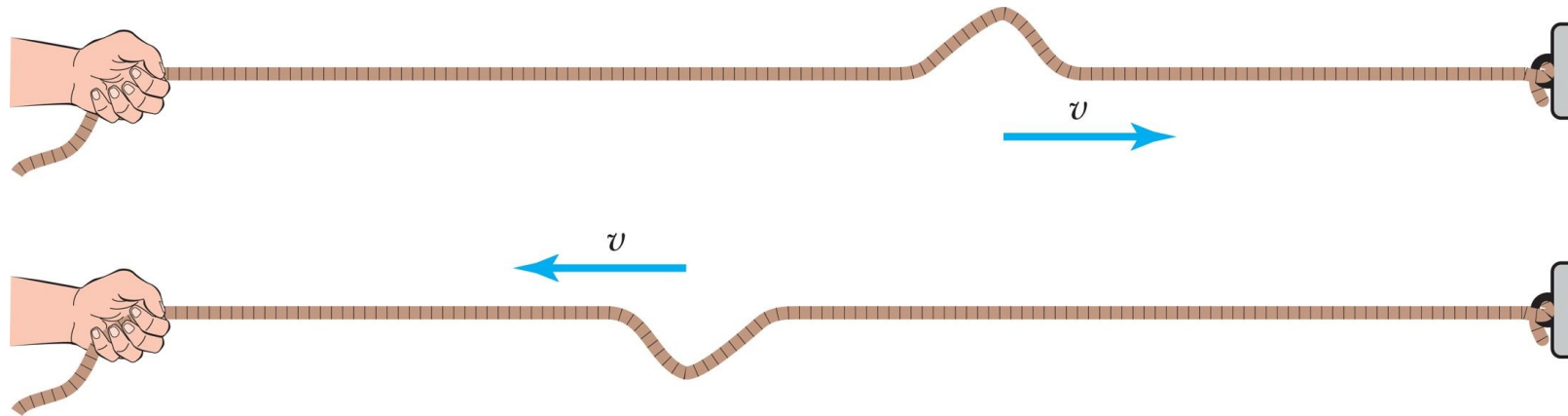


## 6.2 Aspects of Wave Propagation: Wave Fronts and Rays

One inherent aspect of the propagation of waves on a surface or in three dimensions is that the amplitude of the wave necessarily decreases as the wave gets farther from the source.

## 6.2 Aspects of Wave Propagation: Reflection

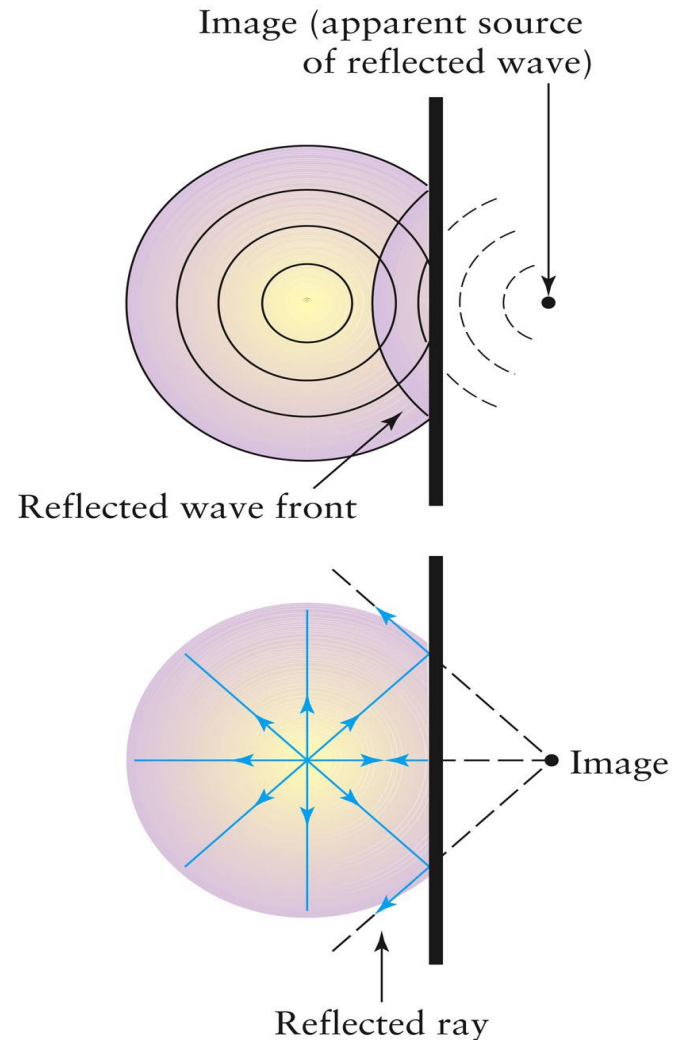
- A **reflection** is when a wave abruptly changes direction.
- A wave is reflected whenever it reaches a boundary of its medium or encounters an abrupt change in the properties of its medium.
- A wave pulse traveling on a rope is reflected when it reaches a fixed. It “bounces” off the end and travels back along the rope.
- Similarly, a wave on a surface or a wave in three dimensions is reflected when it encounters a boundary.





## 6.2 Aspects of Wave Propagation: Reflection

- The wave that “bounces back” is called the *reflected wave*.
- When a wave is reflected from a straight boundary (for surface waves) or a flat boundary (in three dimensions), the reflected wave appears to be expanding out from a point behind the boundary.
- This point is called the image of the original wave source.

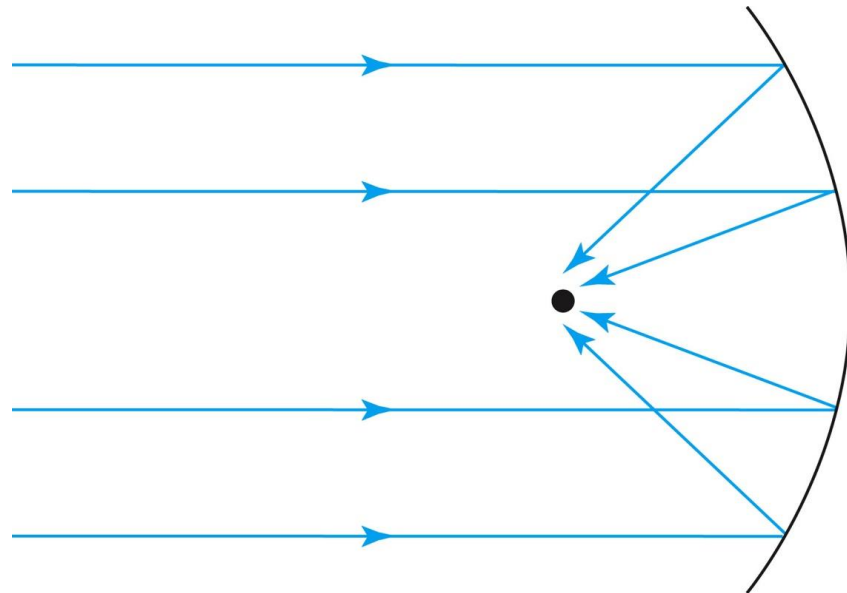


## 6.2 Aspects of Wave Propagation: Reflection

Reflection from surfaces that are not flat (or straight) can cause interesting things to happen to waves.

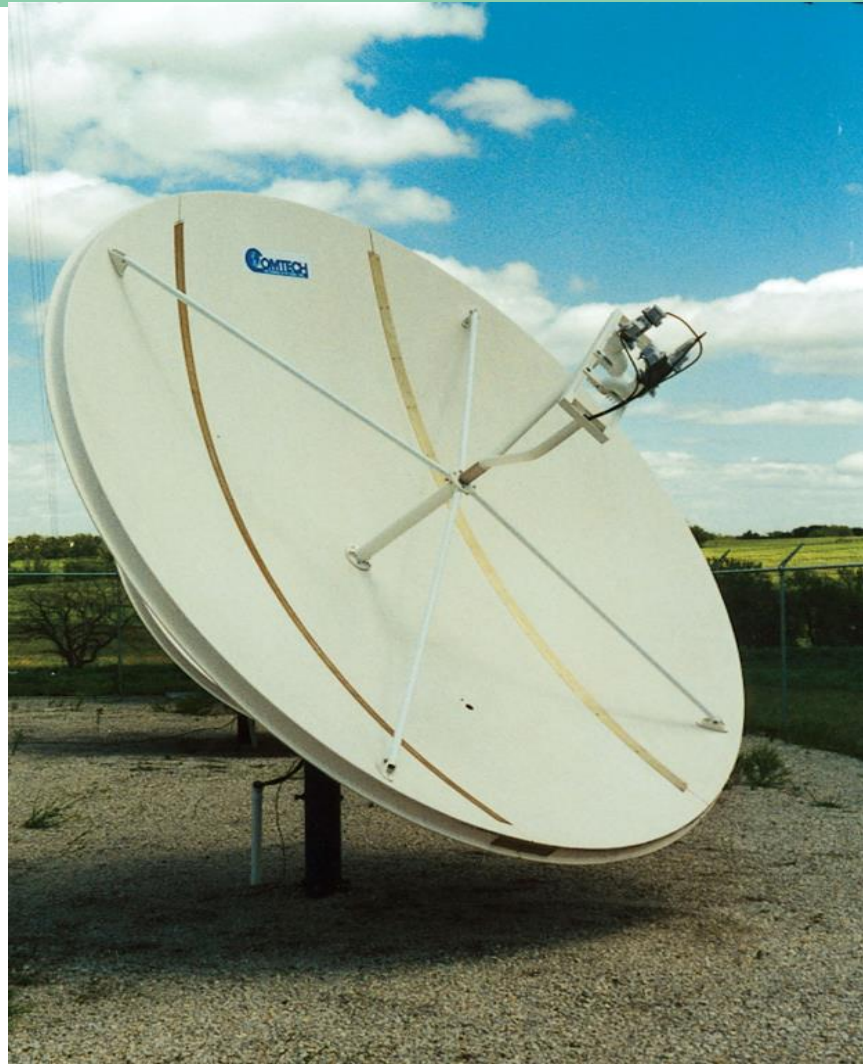
For a curved surface, the reflections can be focused to a point.

Note that the rays representing the reflected part of the wave are converging toward each other. This means that the amplitude of the wave is increasing—the wave is being “focused.”



## 6.2 Aspects of Wave Propagation: Reflection

Parabolic microphones seen on the sidelines of televised football games use this principle to reinforce the sounds made on the playing field. Satellite receiving dishes do the same with radio waves

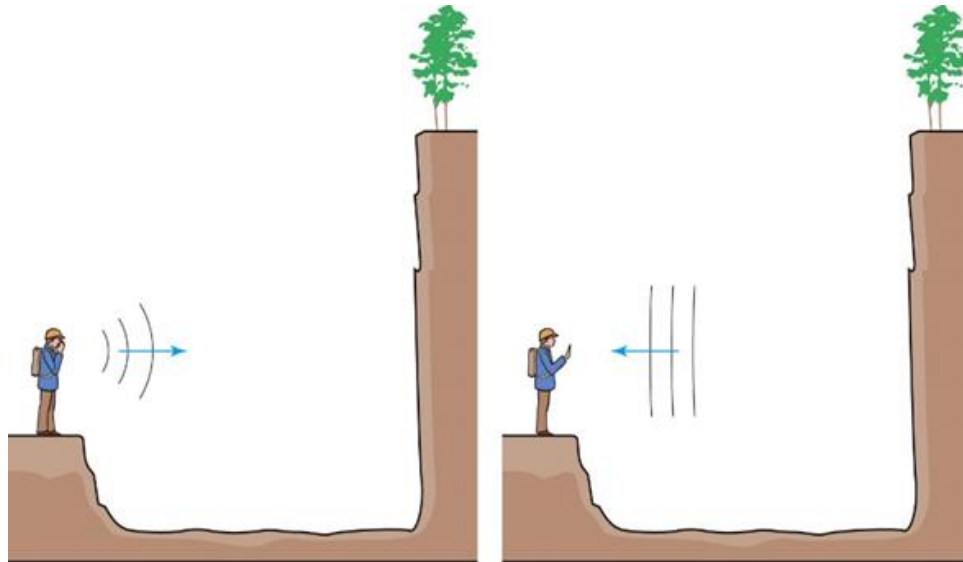


## 6.2 Aspects of Wave Propagation: Reflection

- **Echolocation** is the process of using the waves reflected from an object to determine its location. *Radar* and *sonar* are two examples.
- Basic echolocation uses reflection only:
  - A wave is emitted from a point, reflected by an object of some kind, and detected on its return to the original point.
- The time between the emission of the wave and the detection of the reflected wave (the round-trip time) depends on the speed of the wave and the distance to the reflecting object.

## 6.2 Aspects of Wave Propagation: Reflection

For example, if you shout at a cliff and hear the echo 1 second later, you know that the cliff is approximately 172 meters away. This is because the sound travels a total of 344 meters (172 meters each way) in 1 second (at room temperature). If it takes 2 seconds, the cliff is approximately 344 meters away, and so on.

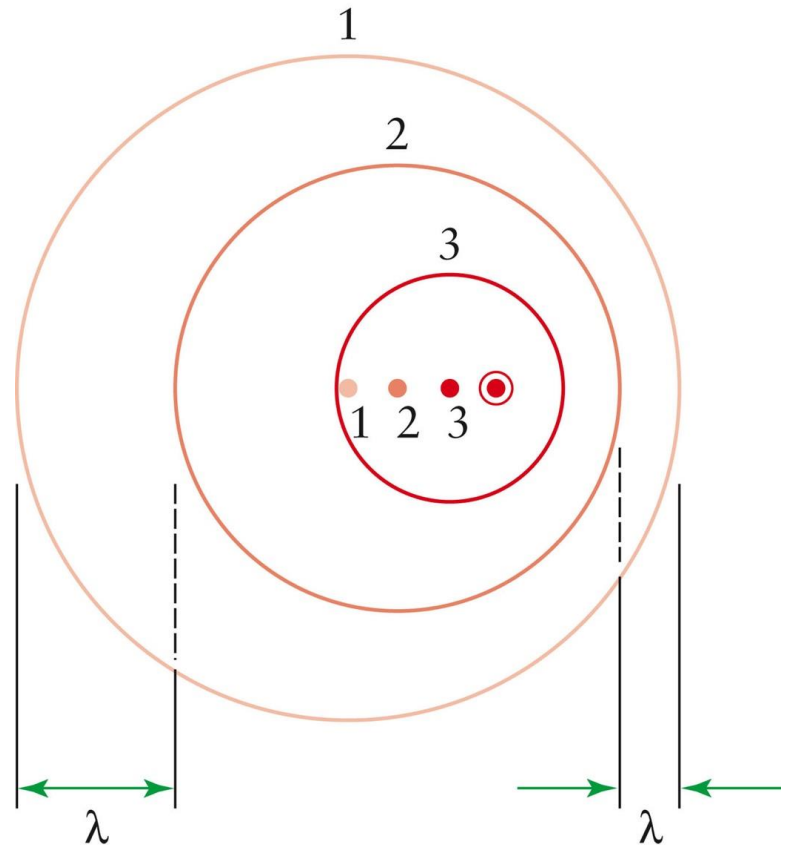


## 6.2 Aspects of Wave Propagation: Doppler Effect

- The ***Doppler effect*** is an apparent change in a wave's wavelength due to the relative motion between the source and receiver.
- Each wave front expands outward from the point where the source was when it emitted that wave front.
- Ahead of the moving source, the wave fronts are bunched together. This means that the wavelength is *shorter* than when the source is at rest, and therefore the frequency of the wave is *higher*.
- Behind the moving source, the wave fronts are spread apart: the wavelength is *longer*, and the frequency is *lower* than when the source is motionless.
- In both places, the higher the speed of the wave source, the greater the change in frequency.

## 6.2 Aspects of Wave Propagation: Doppler Effect

A wave source moves to the right with constant speed. Each dot indicates the source's location when a wave front was emitted. Wave front 1 was emitted when the source was at position 1, and similarly for 2 and 3. Ahead of the source (to the right), the wavelength of the wave is decreased. Behind the source (to the left), the wavelength is increased.



## 6.2 Aspects of Wave Propagation: Doppler Effect

The speed of a wave in a medium is constant and is not affected by any motion associated with the wave source.

Thus, if the wavelength goes up, the frequency must go down, and vice versa, to yield a constant wave speed:

$$v = f \lambda$$



## 6.2 Aspects of Wave Propagation: Shock Waves

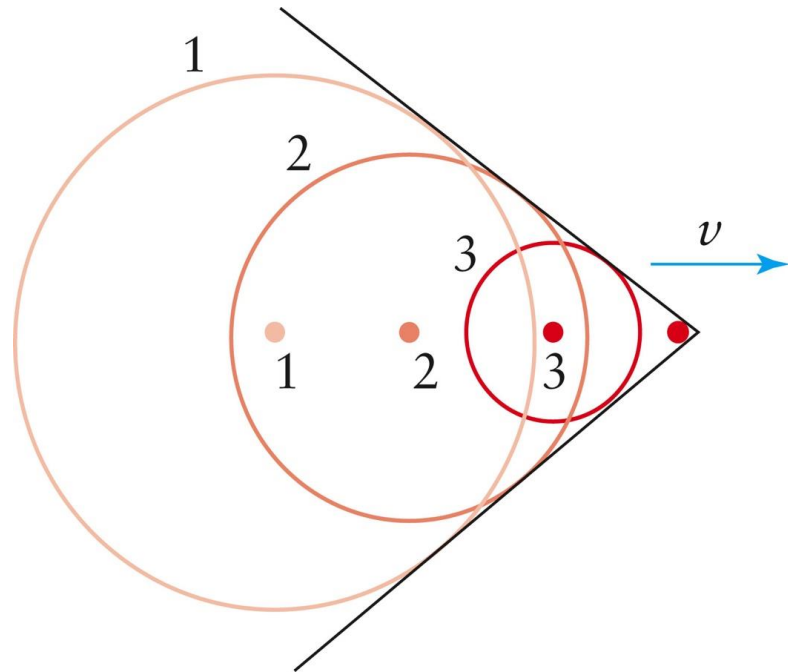
- A **shock wave** occurs whenever the speed of the source is greater than the wave speed.
- The medium cannot respond fast enough to propagate the wave.
  - The crests essentially “pile up” in front of the source.
- This build-up causes a large-amplitude pulse.
  - This produces a sonic boom for supersonic aircraft.
  - A fast boat creates a bow wave.



## 6.2 Aspects of Wave Propagation: Shock Waves

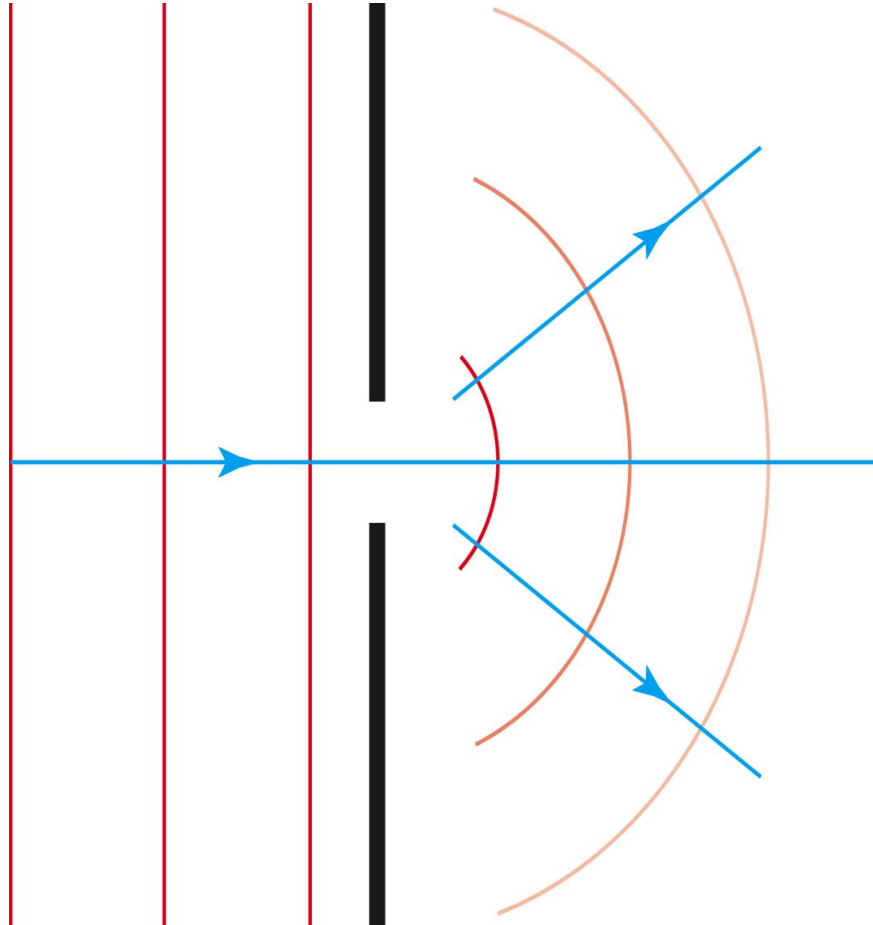
■ Watching the waves expand from the source over time, we can construct a leading edge for the shock wave.

- The angle of this leading-edge was an important discovery for jet-planes.
  - It led to delta-wing configurations.



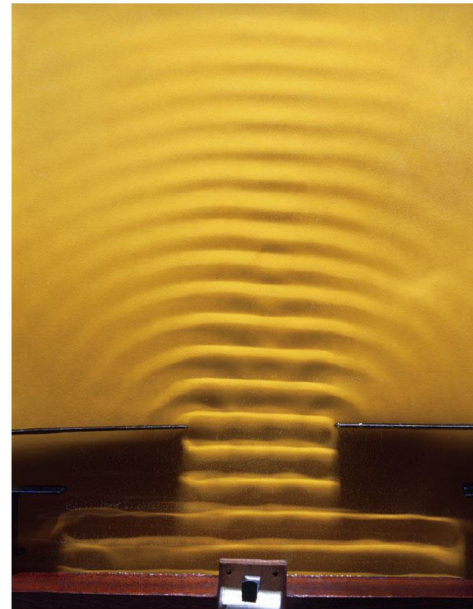
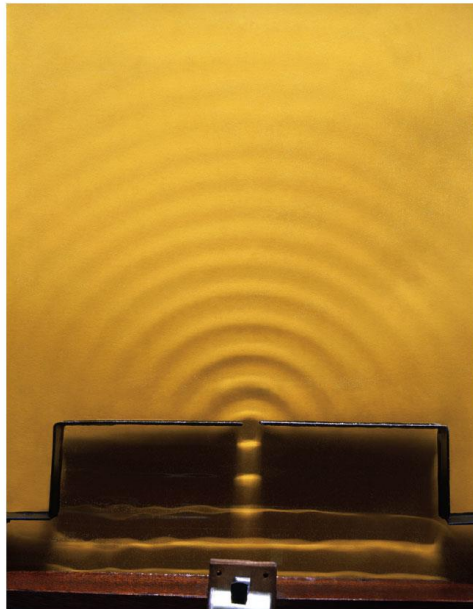
## 6.2 Aspects of Wave Propagation: Diffraction

- **Diffraction** results whenever a wave has to travel past a barrier or obstruction.
- As the wave travels through the opening, the outgoing waves bend.
- The amount of diffraction depends on the wavelength and the size of the obstruction.



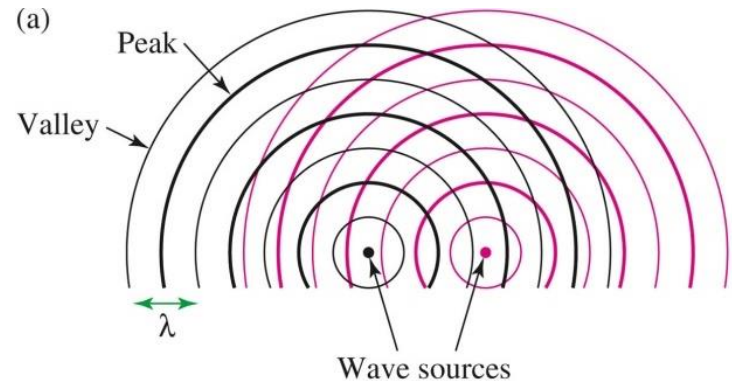
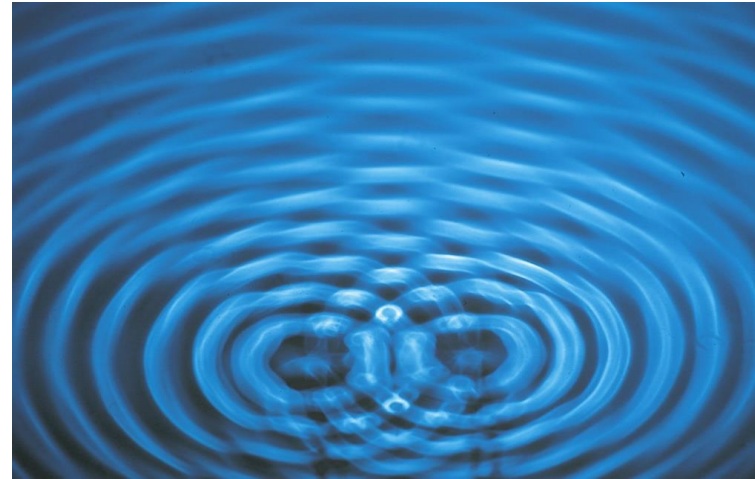
## 6.2 Aspects of Wave Propagation: Diffraction

- When the opening is much larger than the wavelength, there is little diffraction.
- The amount of diffraction increases as the wavelength becomes more similar to the size of the opening.



## 6.2 Aspects of Wave Propagation: Interference

- *Interference* occurs whenever two or more waves overlap
- When the waves interfere to create a larger amplitude, we call it *constructive interference*.
- When the waves interfere to reduce the amplitude, it is called *destructive interference*.



## 6.3 Sound

- Our most common experience with sound is in air, but it can travel in any solid, liquid, or gas.
- when you speak, much of what you hear is sound that travels to your ears through the bones and other tissues in your head.
- The speed of sound in a substance depends on:
  - the mass of its constituent atoms
  - the strength of the forces between the atoms

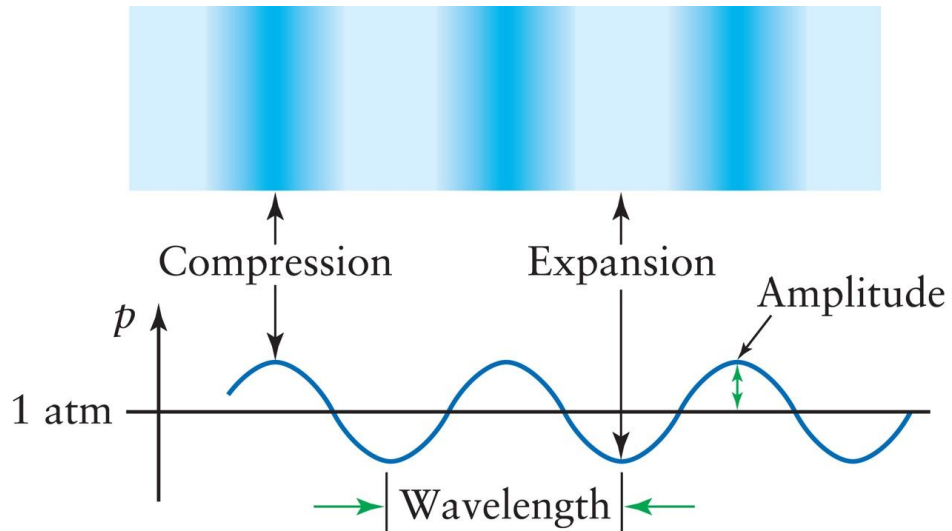
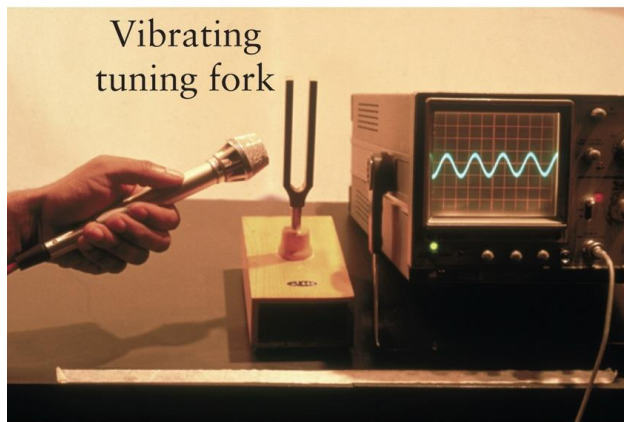
## 6.3 Sound

The speed of sound is generally higher in solids than in liquids and gases because the forces between the atoms and molecules in solids are very strong.

| Substance                                         | Speed* |        |
|---------------------------------------------------|--------|--------|
|                                                   | (m/s)  | (mph)  |
| Air                                               |        |        |
| At -20°C                                          | 320    | 715    |
| At 20°C                                           | 344    | 770    |
| At 40°C                                           | 356    | 795    |
| Carbon dioxide                                    | 269    | 600    |
| Helium                                            | 1,006  | 2,250  |
| Water                                             | 1,440  | 3,220  |
| Human tissue                                      | 1,540  | 3,450  |
| Aluminum                                          | 5,100  | 11,400 |
| Granite                                           | 4,000  | 9,000  |
| Iron and steel                                    | 5,200  | 11,600 |
| Lead                                              | 1,200  | 2,700  |
| * At room temperature (20°C) except as indicated. |        |        |

## 6.3 Sound: Pressure Waves

- Sound is produced by anything that is vibrating and causing the air molecules next to it to vibrate.
- The figure is a representation of a sound wave that was emitted by a vibrating tuning fork. The shading represents the air molecules
- The wave looks very much like a longitudinal wave on a Slinky.
- These compressions and expansions travel at 344 m/s (at room temperature).



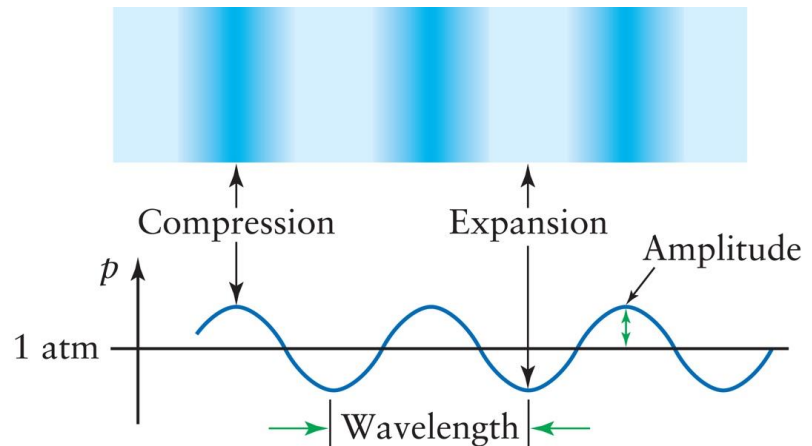


## 6.3 Sound: Pressure Waves

Typically, we represent a sound wave as a transverse wave (even though it is not).

- A region of compression is drawn as a crest.
- A region of expansion is drawn as a trough.

The air pressure in each compression is higher than normal atmospheric pressure because the air molecules are squeezed closer together. Similarly, the pressure in each expansion is below atmospheric pressure.



## 6.3 Sound: Pressure Waves

It is more convenient to think of sound as regular fluctuations of air pressure than as vibrations of molecules, although it is both.

It is these pressure variations that our ears detect and convert into the sensation of sound.

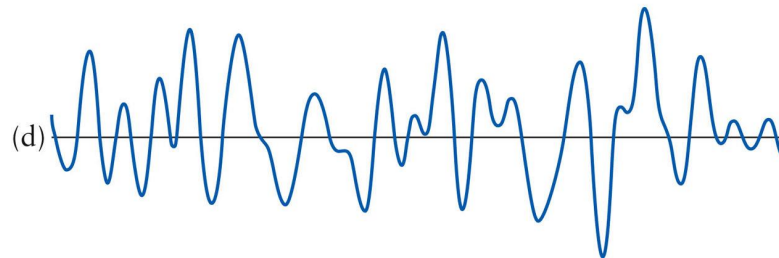
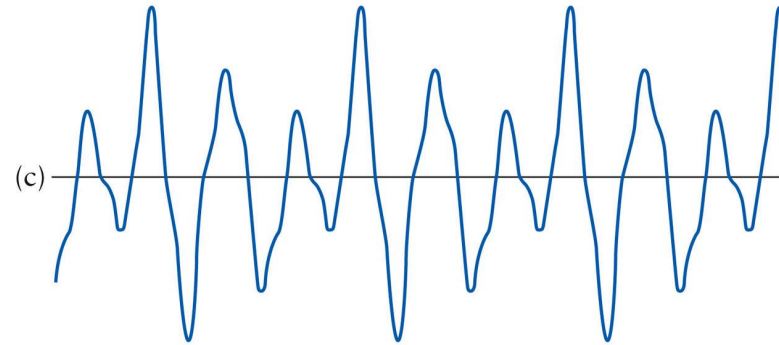
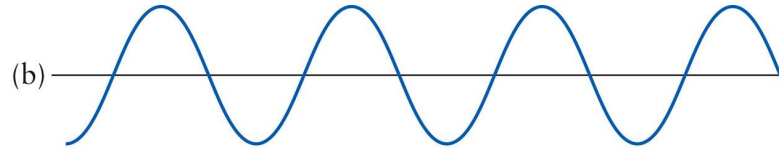
The eardrum is a flexible membrane that responds to pressure changes.

The oscillating pressure of a sound wave forces the eardrum to vibrate in and out.

A remarkable set of physiological structures within the ear converts this oscillation of the eardrum into an electrical signal to the brain that is perceived as sound.

## 6.3 Sound: Pressure Waves

- A **waveform** of a sound wave is a graph of the air-pressure fluctuations caused by the sound wave versus time.
- sounds can be classified by their waveforms:
  - A **pure tone** is a sound with a sinusoidal waveform.
  - A **complex wave** is a sound that is not pure.
  - **Noise** is sound that has a random waveform. It does not have a definite wavelength or period.



## 6.3 Sound: Pressure Waves

- Sound with frequencies below our audible range is called **infrasound** (Below about 20 Hz)
- Sound with frequencies above our audible range is called **ultrasound** (Above about 20,000 Hz)

## 6.3 Sound: Sound Applications

High-frequency ultrasound, typically 3.5 million hertz, is sent into the body and is partially reflected as it encounters different types of tissue. These reflections are analyzed and used to form an image on a video monitor.

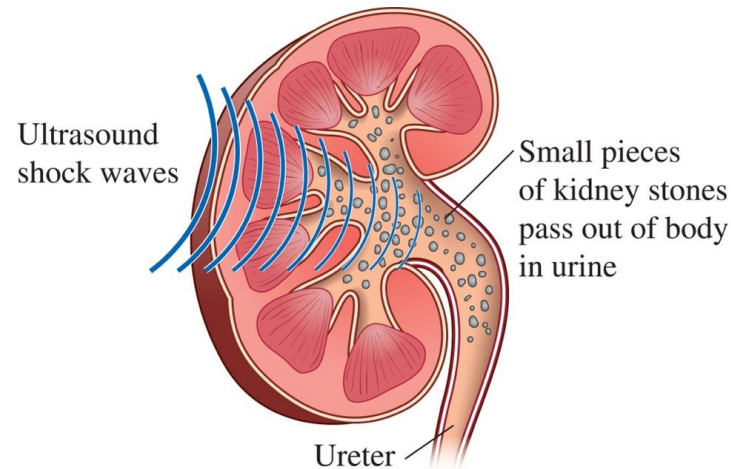
Some sophisticated ultrasonic scanning devices also use the Doppler effect. The beating heart of a fetus and the flow of blood in arteries can be monitored by detecting the frequency shift of the reflected ultrasound.



## 6.3 Sound: Sound Applications

Recently developed *acoustic surgery* uses ultrasound for tasks such as destroying tumors. Focused, high-intensity sound causes heating that destroys tissue. The precision of such an “acoustic scalpel” can exceed that of a conventional knife.

Another use of ultrasound in medicine is *ultrasonic lithotripsy*, a procedure that breaks up kidney stones that have migrated to the bladder. A large-amplitude 23,000- to 25,000-Hz sound wave travels through a steel tube inserted into the body and placed in contact with the stone. The ultrasound breaks the stone into small pieces, somewhat like a singer breaking a wineglass.



# Important Equations

## IMPORTANT EQUATIONS

| Equation                    | Comments                                                            | Equation                        | Comments                                                      |
|-----------------------------|---------------------------------------------------------------------|---------------------------------|---------------------------------------------------------------|
| $\rho = \frac{m}{l}$        | Linear mass density of rope, wire, string, Slinky, etc.             | $v = f\lambda$                  | Relates frequency, wavelength, and speed for continuous waves |
| $v = \sqrt{\frac{F}{\rho}}$ | Speed of waves on linear media (a rope, wire, string, Slinky, etc.) | $v = H_0 d$                     | Hubble relation for expansion of the universe                 |
| $v = 20.1 \times \sqrt{T}$  | Speed of sound waves in air (SI units)                              | amplitude $\propto \frac{1}{d}$ | Dependence of sound amplitude on distance                     |